

NATIONAL UNIVERSITY OF SINGAPORE

EXAMINATION FOR
(Semester I: 2018/2019)

EE5308 – ANTENNA ENGINEERING

November 2018 – Time Allowed: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **THREE (3)** questions, and comprises **SEVEN (7)** printed pages.
2. **Question 1** carries **FIFTY (50) MARKS** and **Questions 2-3** each carries **TWENTY-FIVE (25)** marks, making a total of **ONE HUNDRED (100)** marks for the entire paper.
3. This is an **OPEN-BOOK** examination.

FUNDAMENTAL CONSTANTS:

- μ_0 : free-space permeability = $4\pi \times 10^{-7}$ H/m
 ϵ_0 : free-space permittivity = 8.852×10^{-12} F/m
 c : velocity of light in free space = 3×10^8 m/s
 η_0 : free-space intrinsic impedance = $120\pi \Omega$

Q.1 (Total: 50 marks)

Choose the most SUITABLE answers (only ONE for each question) for the following questions.

- d 1. Compared to a thin straight cylindrical half-wave dipole in free space, a thin straight cylindrical quarter-wave monopole, which is normally positioned above an infinite perfect electrically conducting ground plane, features
- (a) the same radiation pattern as the dipole *half*
 - (b) the same directivity $D_{\text{mono}} = 2D_{\text{dipole}}$
 - (c) the same gain
 - (d) half input impedance $Z_{\text{mono}} = Z_{\text{dipole}}/2$

(3 marks)

- a 2. A shorting-pin is used in the feeding end of an electrically low-profile inverted-L antenna to form an inverted-F antenna for
- (a) improving the impedance matching of the inverted-L antenna
 - (b) shortening the radiating arm of the inverted-L antenna
 - (c) enhancing mechanical robustness of the inverted-L antenna
 - (d) further lowering the profile of the inverted-L antenna

(3 marks)

- b 3. When “dBi” is used in the expression of a maximum gain of an antenna, it suggests that the gain is the ratio of radiation intensity of the antenna to
- (a) a half-wave dipole. *dBd*
 - (b) an isotropic radiator.
 - (c) an omni-directional monopole antenna.
 - (d) the antenna itself.

(3 marks)

- a 4. If the reflection coefficient at the input of an antenna is 0.5, VSWR and return loss of the antenna are:
- (a) 3 and 6 dB; $\Gamma = 0.5$
 - (b) 3 and -6dB; $\rho = \frac{1+\Gamma}{1-\Gamma} = 3$
 - (c) 3 and 2 dB
 - (d) 2 and -2 dB $RL = -20 \log|\Gamma| = 6$

(3 marks)

a 5. A 5-GHz WiFi indoor access point is installed in the living room. Which one includes the enough information for us to design the antenna fed by a 50-ohm coaxial cable for the access point?

- (a) Frequency range, impedance matching, gain, polarization, and radiation patterns
 (b) Frequency range, impedance matching, gain, and size limitation
 (c) Frequency range, impedance matching, gain, polarization, and size limitation
 (d) Frequency range, gain, polarization, radiation patterns, and size limitation

(3 marks)

b 6. Which antenna will be the MOST possible candidate for a point-to-point narrow-band wireless link operating with high gain?

- (a) A half-wave dipole. $D = 1.64$ or 2.15 dBi
 (b) A Yagi-Uda antenna. $G = 6.14 \text{ dB}$ High Gain + Narrow Band
 (c) A log-periodic dipole antenna. wideband
 (d) A loop antenna. Similar to Hertzian Dipole, $D_{\max} = 3/2 = 1.76 \text{ dBi}$ at $\theta = 90^\circ$

(3 marks)

$$P_r(\text{dB}) = G(\text{dB}) + P_t(\text{dBW}) + 20 \log \frac{\lambda}{4\pi r}$$

7. Using the three-antenna gain method, a standard antenna of known gain of 10 dBi at its operating frequency receives the power of P_s in its boresight when a transmit antenna transmits a power P_t at the far-zone of each other. With the same polarization as the standard antenna, the antenna under test receives the power of $0.5P_s$ in its boresight when the power transmitted by the transmitter is kept P_t . So, the boresight gain of the antenna under test is

- (a) 7 dBi
 (b) 8 dBi
 (c) 10 dBi
 (d) 12 dBi.

$$P_s(\text{dBW}) = 10 \log \frac{P_s}{W} \quad P_s'(\text{dBW}) = 10 \log \frac{P_s}{2W}$$

$$= P_s 10 \log \frac{P_s}{W} + 10 \log \frac{1}{2} \quad \left(10 \log \frac{P_s}{W} \right) - 3 \text{ dB}$$

$$G'(\text{dB}) = P_s'(\text{dBW}) - P_t(\text{dBW}) - K$$

$$= P_s(\text{dBW}) - 3 \text{ dB} - P_t(\text{dBW}) - K$$

(3 marks)

$$G(\text{dB}) = P_r(\text{dBW}) - P_t(\text{dBW}) - K$$

8. If the dipole of directivity of 3.0 dBi has the radiation resistance of 40Ω and the material ohmic loss of 3Ω . The dipole antenna is fed by a 50- Ω feeding line without any impedance matching network between the dipole and feeding line. So, the radiation efficiency and realized gain are, respectively

- (a) 93.0% and 3.0 dBi
 (b) 92.5% and 3.0 dBi.
 (c) 93.0% and 2.67 dBi.
 (d) 92.5% and 2.67 dBi.

$$e_{\text{rad}} = \frac{R_{\text{rad}}}{R_{\text{loss}} + R_{\text{rad}}} = \frac{40}{43} = 93\%$$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{43 - 50}{43 + 50} = -0.075$$

(3 marks)

$$e_m = 1 - \Gamma^2 = 0.994$$

$$e_{\text{ant}} = e_m \cdot e_{\text{rad}} = 0.994 \times 0.93 = 92.4\%$$

$$G = e_{\text{ant}} \cdot D = 0.924 \times 3 = 2.77 \text{ dBi}$$

9. The input impedance of an HF RFID tag antenna (coil) is designed for
- matching to a standard feed-line impedance of 50Ω .
 - matching to the output impedance of the RFID tag chip used.
 - matching to the conjugate output impedance of the RFID tag chip used.
 - high radiation efficiency of the coil antenna.

(3 marks)

10. Consider a full-wave dipole fed by a voltage source at its center as shown in Figure Q1-1. There are three electric current distributions as shown in Figures Q1-1(b-d). Which answer is correct?

- All the three current distributions
- The electric current distribution shown in Figures Q1-1 (b);
- The electric current distribution shown in Figures Q1-1 (c);
- The electric current distribution shown in Figures Q1-1 (d);

(3 marks)

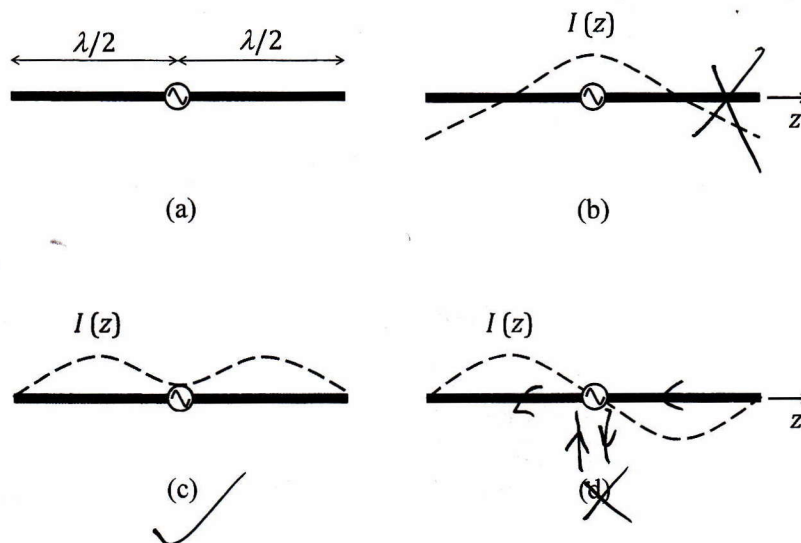


Figure Q1-1

11. A log-periodic dipole antenna (Figure Q.1-2) with optimum parameters is designed to cover the frequency range of 47–80.6 MHz. Suppose the desired gain is 9dBi. Ignore the variation of feed gap, s_n .

- (a) Determine the spacing and scaling factors $\sigma = \underline{0.158}$ and $\tau = \underline{0.862}$ for the optimum design.

(5 marks)

- (b) Determine the minimum number of the dipoles, $N = \underline{5}$ for the realized gain over the desired frequency range.

(5 marks)

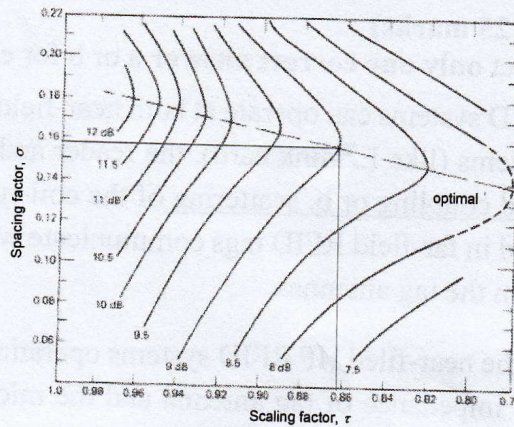
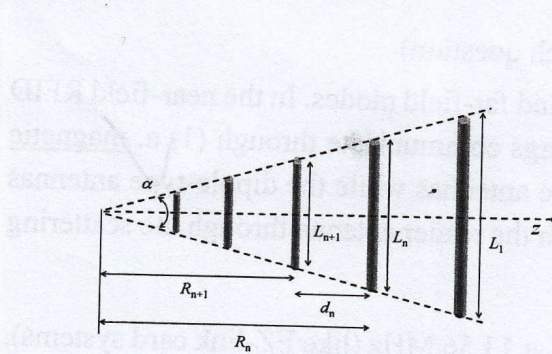
- (c) Tabulate the lengths of all dipoles, L_n in meter and all the element spacing, d_n in meter

(5 marks)

- (d) Select the most suitable feeding scheme for the high-gain design: B

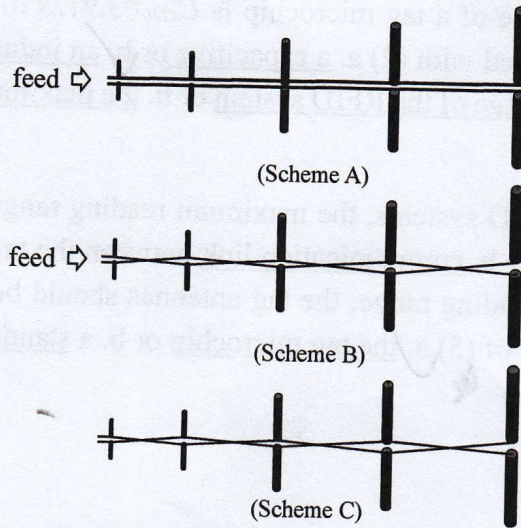
(5 marks)

Hint: to determine optimum design factors from the attached figure.



$$\tau = 0.863$$

$$\sigma = 0.158$$



$$L_1 = 0.5 \lambda_1 = 0.5 \times \frac{c}{f_{min}} = 3.19 \text{ m}$$

$$L_N = 0.5 \lambda_N = 0.5 \times \frac{c}{f_{max}} = 1.86 \text{ m}$$

$$L_{n+1} = \tau L_n$$

$$d_n = 2\sigma L_n$$

$$d_n = 26 L_n$$

$$= 2 \times 0.158 \times L_n$$

Figure Q.1-2

Table Q.1 List of L_n and d_n

	n=1	n=2	n=3	n=4	n=5	n=6
L_n (m)	3.19×10^{-3}	2.75×10^{-3}	2.37×10^{-3}	2.04×10^{-3}	1.76×10^{-3}	✓
d_n (m)	1.01×10^{-3}	0.87×10^{-3}	0.75×10^{-3}	0.64×10^{-3}	0.56×10^{-3}	✓

n	1	2	3	4	5	6
L_n (m)	3.19	2.75	2.37	2.04	1.76	1.51
d_n (m)	1.01	0.87	0.75	0.64	0.56	

Q2. (Total 25 marks)Q.2a: (Select **only one correct answer** a or b for each question)

- (a) RFID systems can operate at both near-field and far-field modes. In the near-field RFID systems (like EZ-link card), the reader and tags communicate through (1) a. magnetic field coupling or b. scattering of the coil-type antennas while the dipole-type antennas used in far-field RFID tags communicate with the reader antenna through the scattering from the tag antennas.
- (b) In the near-field HF RFID systems operating at 13.56 MHz (like EZ-link card systems), the impedance of tag antenna and the microchip are respectively Z_{tag} and Z_{chip} . If the overall output capacitance of a tag microchip is $C_{\text{chip}} = 3.912 \times 10^{-11}$, the coil of the tag antenna should be designed with (2) a. a capacitive or b. an inductive reactance for (3) a. a maximum reading range of the RFID system or b. the maximum radar cross section (RCS) of the antenna.
- (c) In the far-field UHF RFID systems, the maximum reading range is determined by (4) a. power charging link or b. communication link between the tag and reader antennas. To achieve maximum reading range, the tag antennas should be conjugately matched to the output impedance of (5) a. the tag microchip or b. a standard impedance such as 50 ohm.

(3×5 marks)

Q.2b. The straight wire monopole antenna is positioned particular to a perfect electrically conducting plane and fed at the end by a voltage source V_{in} as shown in Figure Q2b.

1. Draw the equivalent dipole antenna of the original monopole antenna using Image Theory with the indication of the source voltages and currents I_{in} along the both original monopole and its image.
2. If a real dipole Ant 1 has exactly the same structure of the equivalent dipole of the original monopole Ant 2 and is fed by voltage source V_{in} , show the gain of a real dipole Ant 1 is one-half of the monopole Ant 2 and the radiation resistance of the dipole Ant 1 is twice of the monopole Ant 2.

For Ant 2: $D = \frac{U_{\text{max}}}{U_{\text{ave}}}$

(5+5 marks)

U_{max} is twice of Ant 1, but U_{ave} is same as Ant 1: $D_2 = 2D_1$
 $\Rightarrow G_1 = \frac{1}{2} G_2$

I_0 is the same as Ant 1, but $V_{\text{in}}/2$

$$Z_2 = \frac{V_{\text{in}}}{I_{\text{in}}} = \frac{V_{\text{in}}/2}{I_{\text{in}}} = \frac{1}{2} Z_1 \quad Z_1 = 2Z_2$$

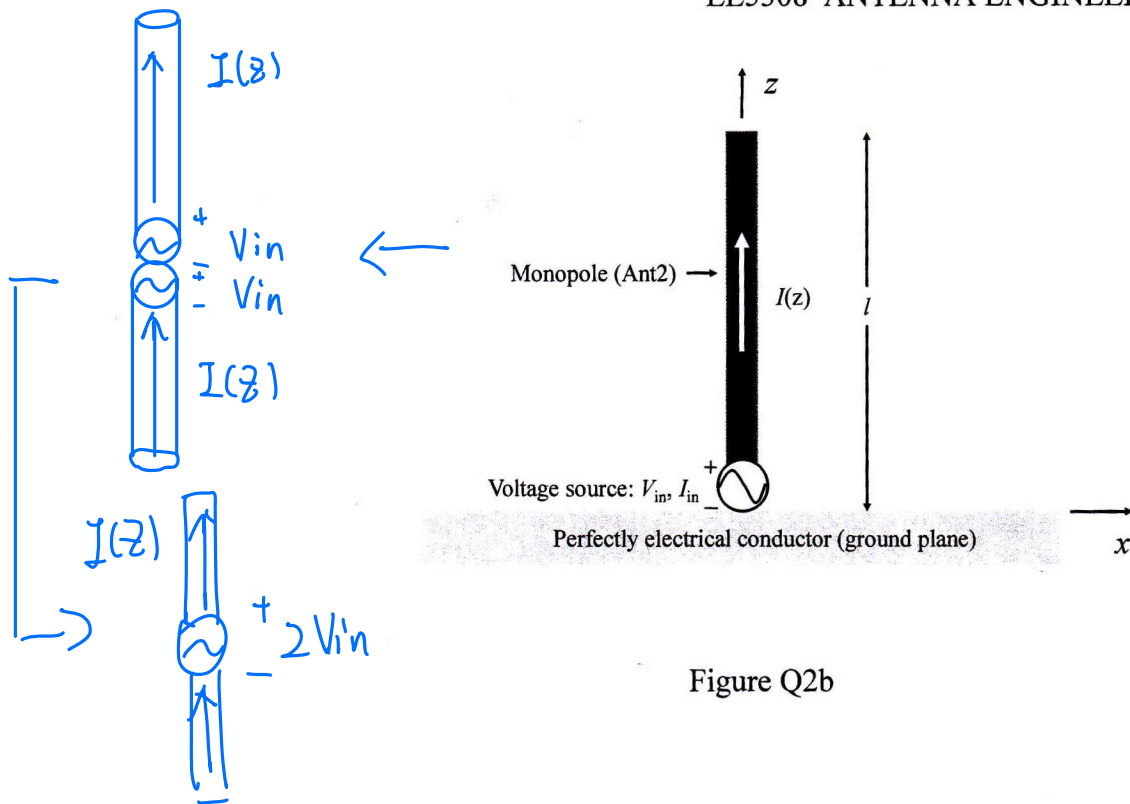


Figure Q2b

Q3. (Total 25 marks)

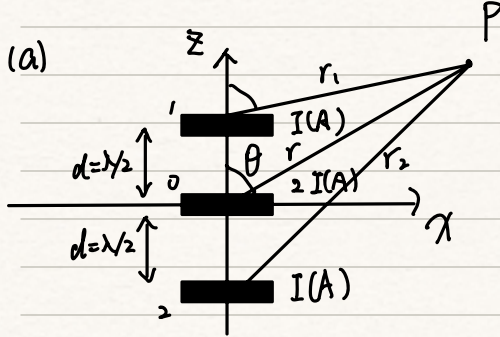
A linear array consisting of three identical infinitesimal dipole antennas of length of l is positioned along the z -axis where all three dipoles are in parallel and pointed to x -direction. The centre dipole is located at the origin. The current excitation for the centre antenna has a magnitude of $2I(A)$ and the adjacent dipoles have the magnitude of $I(A)$. The uniform antenna spacing is $d = \lambda/2$ (half a wavelength) apart.

Hint:

The electrical far-field radiated at an observation point (at a distance $r \gg \text{wavelength}$ from the origin) by an infinitesimal dipole located at the origin carrying the excitation current magnitude of $\mathbf{C} = \mathbf{a}_\theta j \eta \frac{k l l}{4\pi} \frac{e^{-jkr}}{r} |\cos\theta|$.

- Give the total electrical far-field in x - z planes radiated by this array (5 marks)
- Find and simplify the expression of the array factor for this array. (5 marks)
- Find the angles (in degree) for all the possible nulls in the array factor (5 marks)
- Compute the angles of all the maxima in the array factor (5 marks)
- Find the angles (in degree) for all the possible nulls in this array (5 marks)

END OF PAPER



$$\vec{E} = \hat{a}_\theta j\eta \frac{kIL}{4\pi} \frac{e^{-jkr}}{r} |\cos\theta|$$

For amplitude: $r \approx r_1 \approx r_2$

For phase: $r_1 = r - d\cos\theta$, $r_2 = r + d\cos\theta$

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

$$= \hat{a}_\theta \frac{j\eta kIL}{4\pi r} |\cos\theta| e^{-jkr_1} + \hat{a}_\theta \frac{j\eta k \cdot 2I \cdot l}{4\pi r} |\cos\theta| e^{-jkr} + \hat{a}_\theta \frac{j\eta kIL}{4\pi r} |\cos\theta| e^{-jkr_2}$$

$$= \hat{a}_\theta \frac{j\eta kIL}{4\pi r} |\cos\theta| (e^{-jkr_1} + 2e^{-jkr} + e^{-jkr_2})$$

$$= \hat{a}_\theta \frac{j\eta kIL}{4\pi r} |\cos\theta| e^{-jkr} (e^{jkd\cos\theta} + 2 + e^{-jkd\cos\theta})$$

$$= \hat{a}_\theta \frac{j\eta kIL}{4\pi r} |\cos\theta| e^{-jkr} \cdot 2[\cos(kd\cos\theta) + 1]$$

$$\frac{2\eta}{\lambda} \times \frac{\lambda}{2}$$

(b) Array Factor: $F_A(\theta) = 2[\cos(kd\cos\theta) + 1] = 2[\cos(\pi\cos\theta) + 1]$ $0 \leq \theta \leq \pi$

(c) Nulls in AF: $\cos(\pi\cos\theta) = -1 \Rightarrow \pi\cos\theta = \pi + 2n\pi \Rightarrow \cos\theta = 1 + 2n \Rightarrow \theta = 0^\circ, 180^\circ$

(d) Maxima in AF: $\cos(\pi\cos\theta) = 1 \Rightarrow \pi\cos\theta = 2n\pi \Rightarrow \cos\theta = 2n \Rightarrow \theta = 90^\circ$

(e) Null in this array: $|\cos\theta| = 0, \Rightarrow \theta = 90^\circ$
Then: $\theta = 0^\circ, 90^\circ, 180^\circ$

NATIONAL UNIVERSITY OF SINGAPORE

EXAMINATION FOR
(Semester I: 2020/2021)

EE5308 – ANTENNA ENGINEERING

November/December 2020 – Time Allowed: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **Three (3)** sets of questions and comprises **SIX (6)** printed pages.
2. **ALL Questions are compulsory for all students.**
3. Question 1 each carries FIFTY (50) marks and each of Questions 2 and 3 carries TWENTY-FIVE (25) marks, making a total of **ONE HUNDRED (100) marks** for the entire paper.
4. This is an **OPEN-BOOK** examination.

FUNDAMENTAL CONSTANTS:

- μ_0 : free-space permeability = $4\pi \times 10^{-7}$ H/m
 ϵ_0 : free-space permittivity = 8.852×10^{-12} F/m
 c : velocity of light in free space = 3×10^8 m/s
 η_0 : free-space intrinsic impedance = $120\pi \Omega$

Q.1 (Total: 50 marks)

Choose the correct answers (only one for each question) for the following questions.

1. The radiation of electromagnetic waves is generated by
 - (a) steady electric currents without acceleration.
 - (b) linearly time-varying currents without acceleration.
 - (c) alternating currents with acceleration.

(3 marks)

2. An electrically short dipole and an electrically small loop,
 - (a) have the same normalized power radiation pattern.
 - (b) have the same radiation impedance
 - (c) have the same wave impedance in a near-field zone as the distance from the antenna is $r < 1/(2\pi)$

(3 marks)

3. A shorting-pin is used in an electrically low-profile inverted-L antenna to form an inverted-F antenna for
 - (a) further lowering the profile of the inverted-L antenna
 - (b) improving the impedance matching of the inverted-L antenna
 - (c) shortening the radiating arm of the inverted-L antenna

(3 marks)

4. To improve the impedance matching performance of a two-armed Archimedean spiral antenna fed at its starting ends (with the smallest radii), we usually introduce:
 - (a) an absorbing loading at the ends of the spirals;
 - (b) a cavity to back the spiral antenna;
 - (c) an impedance matching network at the starting ends;

(3 marks)

5. If the VSWR at the input of the antenna is 2:1, the return loss of an antenna is:
 - (a) 9.5 dB;
 - (b) -9.5 dB
 - (c) 14 dB;

(3 marks)

6. The broad bandwidth of *directive* antennas suggests that the antenna can operate with
 - (a) well impedance matching over a desired bandwidth
 - (b) consistent achieved gain over a desired bandwidth
 - (c) unchanged polarization over a desired bandwidth
 - (d) all above

(3 marks)

7. Which antenna will be the MOST possible candidate for a wireless system operating with a high gain over a wide bandwidth for a wide coverage?
- A half-wave dipole.
 - A log-periodic dipole antenna.
 - A Yagi-Uda antenna.
 - A phased/scanning antenna array.
- (3 marks)
8. Indicate the only one CORRECT statement as below
- In the near-field 13.56 MHz-HF RFID systems, the reader and tag antennas can be either dipole-types or coil-types.
 - In the passive RFID systems, the tag antennas are used to harvest the power from the readers and communicate ID information between the tags and readers.
 - To achieve a maximum reading range, the RFID tag antenna should be conjugate matched to a standard impedance of 50Ω .
- (3 marks)
9. Two straight thin-wire half-wave dipoles A and B fed by the current with the same amplitude are used to generate circular polarization in z-direction. Dipole A is located at origin O in x -direction. How to position and excite Dipole B?
- At origin O in y -direction and fed by current with 90° phase shift to Dipole A.
 - At origin O in y -direction and fed by current with 0° phase shift to Dipole A.
 - At origin O in z -direction and fed by current with 90° phase shift to Dipole A.
- (3 marks)
10. To achieve broadband and high gain, which feeding schemes of a log-periodic antenna array (see figure as below) are correct (select only ONE answer)
- Schemes (1) and (2)
 - Schemes (2) and (3)
 - Schemes (1) and (4)
 - Schemes (2) and (4)

(3 marks)

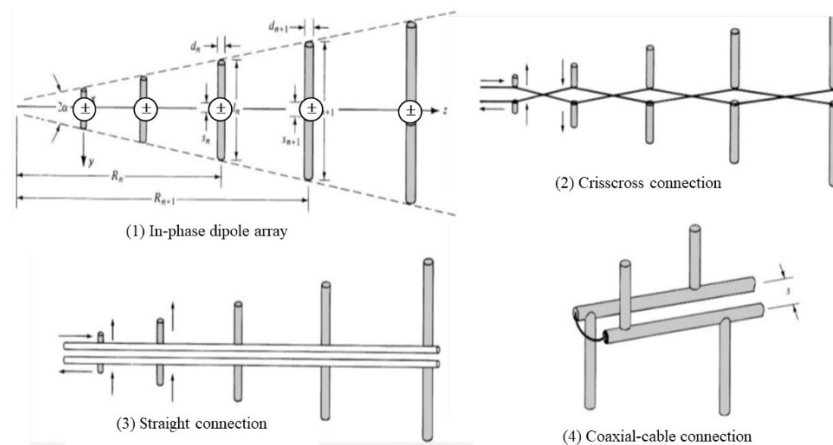


Figure 1.1

11. In a microwave link operating at 8 GHz, the distance is 5 km from a transmitting antenna of gain of 30 dBi to a receiving antenna of gain of 35 dBi. Polarization mismatching loss between transmitting and receiving antennas is 1 dB. What is the received power over the path if the transmitted power is 1 KW?

(a) -30.48 dBW
 (b) -40.48 dBW
 (c) -59.48 dBW
 (d) -60.48 dBW

(5 marks)

12. Using the method of three-antenna gain method, a standard antenna of known gain of 10 dBi at its frequency receives the electrical power of P_s when a transmit antenna transmits a power P_t at its far field. With the same polarization as the standard antenna, the antenna under test receives the electrical power of $2P_s$ when the transmitted power is kept P_t . So, the gain of the antenna under test is

(a) 8 dBi
 (b) 10 dBi.
 (c) 12 dBi.
 (d) 13 dBi.

(5 marks)

13. If the dipole of directivity of 3 dBi has the radiation resistance of 40Ω and the material ohmic loss of 3Ω . The dipole antenna is matched to a $50\text{-}\Omega$ feeding line using an impedance matching network between the dipole and feeding line. The impedance matching network introduces the additional ohmic loss of 5Ω . If the return loss between the output of the impedance network and input of the antenna is 10 dB, So, the antenna efficiency and achieved gain of the antenna are

(a) 83.3% and 2.92 dBi.
 (b) 75.0% and 2.88 dBi.
 (c) 88.9% and 2.95 dBi.
 (d) 93.0% and 2.88 dBi.

(5 marks)

14. A square thin-wire loop with circumference of one wavelength is fed at the center of its bottom (point a), which sketch of current distribution (in Figure 1.2) in loop is correct?

(a) (A)
 (b) (B)
 (c) (C)

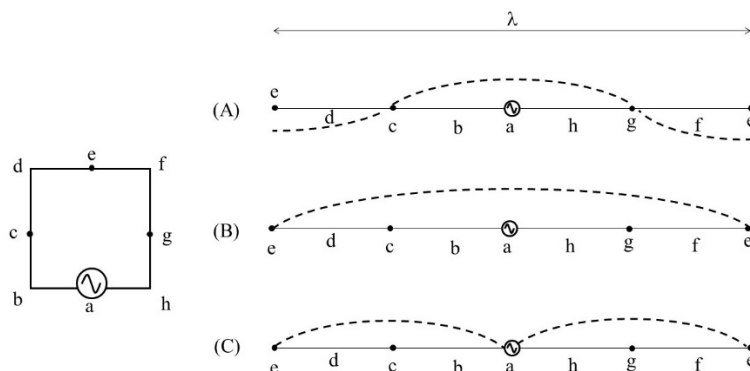


Figure 1.2

(5 marks)

Q.2 (Total 25 marks)

A linear array consisting of three identical infinitesimal dipole antennas of length of l is positioned along the z -axis where all three dipoles are in parallel and pointed to x -direction. The centre dipole is located at the origin. The current excitation for the centre antenna has a magnitude of $2I$ (A) and the adjacent dipoles have the magnitude of I (A). The uniform antenna spacing is $d=\lambda$ (one wavelength) apart.

Hint:

The electric far-field radiated at an observation point (at a distance $r \gg \text{wavelength}, \lambda$, from the origin) by an infinitesimal dipole located at the origin carrying the excitation current magnitude of I :

$$\mathbf{E} = \mathbf{a}_\theta j \eta \frac{k l I}{4\pi} \frac{e^{-jkr}}{r} |\cos\theta|.$$

- Give the total electrical far-field in x - z planes radiated by this array (5 marks)
- Find and simplify the expression of the array factor for this array. (5 marks)
- Find the angles (in degree) for all the possible nulls in the array factor (5 marks)
- Compute the angles of all the maxima in the array factor (5 marks)
- Find the angles (in degree) for all the possible nulls in this array (5 marks)

Q3. (Total 25 marks)

A square array consisting of four isotropic antenna elements 1-4 is configured as shown in Fig Q3. All the elements are equally excited with a magnitude of I (A) and no phase difference. The uniform antenna spacing is $d=\lambda/2$ (a half wavelength) apart.

Hint:

The radiated electric field at an observation point at a far-field zone (at a distance $r \gg \lambda$ from the origin) by an isotropic source located at the origin carrying the excitation current magnitude of I : $E \propto I_0 \frac{e^{-jkr}}{r}$.

- Give all the approximate distances r_1 and r_2 in terms of r_0 , spacing d and angle θ at the far-field observation point (5 marks)
- Find and simplify the expression of the array factor in the y-z plane for this array. (10 marks)
- Find angles (in degree) for all the possible nulls in the array factor in the y-z plane. (5 marks)
- Find all the maxima and their angles (in degree) in the array factor in the y-z plane. (5 marks)

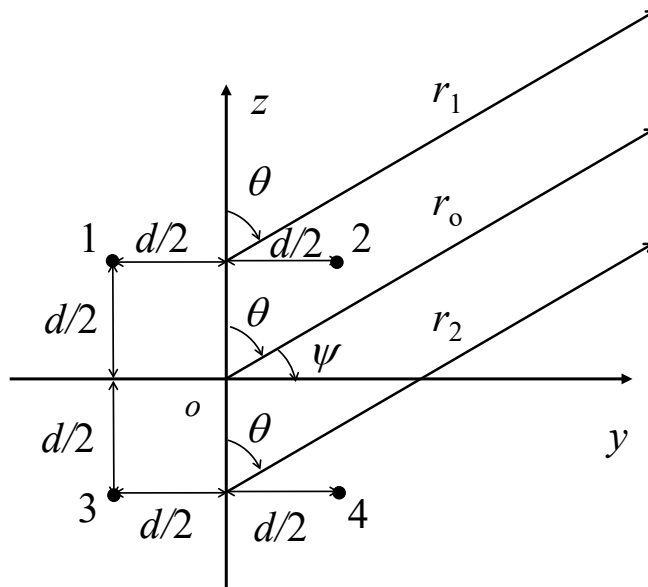


Fig Q3

NATIONAL UNIVERSITY OF SINGAPORE

EXAMINATION FOR
(Semester I: 2022/2023)

EE5308 – ANTENNA ENGINEERING

November/December 2022 – Time Allowed: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **FOUR (4)** questions, and comprises **SEVEN (7)** printed pages.
2. A total of **ONE HUNDRED (100=30+30+20+20)** marks for the entire paper.
3. This is an **OPEN-BOOK** examination.
4. Write **ALL** answers on **ANSWER BOOK**

FUNDAMENTAL CONSTANTS:

- μ_0 : free-space permeability = $4\pi \times 10^{-7}$ H/m
 ϵ_0 : free-space permittivity = 8.852×10^{-12} F/m
 c : velocity of light in free space = 3×10^8 m/s
 η_0 : free-space intrinsic impedance = 120π Ω



Q.1 (Total: 30 marks)

Choose the most SUITABLE answers (only ONE for each question) for the following questions.

- b 1. Compared to a half-wave straight dipole in free space, a straight quarter-wave monopole above an infinite ground has
- (a) a different radiation pattern above the ground plane
 - (b) a doubled directivity
 - (c) the same input impedance

(3 marks)

- d 2. Which antenna was NOT used in Heinrich Rudolf Hertz's experiments of electromagnetic waves?
- (a) Dipole
 - (b) Horn
 - (c) Loop
 - (d) Microstrip antenna

(3 marks)

- a 3. A helix antenna operates at 100 MHz. Its circumference of (one turn) helix is 3 m and spacing between turns 0.8 m. The helix antenna is operating at its
- (a) axial mode
 - (b) normal mode
 - (c) both axial and normal modes

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{100 \times 10^6} = 3 \text{ m}$$

(3 marks)

- a 4. If the reflection coefficient at the input of an antenna is 0.5, VSWR and return loss of the antenna are:
- (a) 3 and 6 dB;
 - (b) 3 and -6dB;
 - (c) 3 and 2 dB
 - (d) 2 and -2 dB

$$|\Gamma| = 0.5$$

$$\text{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|} = 3$$

$$RL = -20 \log 0.5 = 6 \text{ dB}$$

(3 marks)

- c a (5.) A 5-GHz WiFi indoor access point is installed in the living room of a flat. Which one has the enough information for us to design the antenna fed by a 50-ohm coaxial cable for the access point?
- (a) Frequency range, impedance matching, gain, polarization, and radiation patterns
 - (b) Frequency range, impedance matching, gain, and size limitation
 - (c) Frequency range, impedance matching, gain, polarization, and size limitation
 - (d) Frequency range, gain, polarization, radiation patterns, and size limitation X
 - (e) Impedance matching, gain, polarization, radiation patterns, and size limitation X

(3 marks)



6. Which antenna will be the MOST possible candidate for a point-to-point narrow-band wireless link operating with high gain?

- (a) A half-wave dipole. $\times$
- (b) A large-scale antenna array.
- (c) A log-periodic dipole antenna. $\times$
- (d) A loop antenna. $\times$
- (e) A phased/scanning antenna array.

(3 marks)

7. Using the three-antenna gain method, a standard antenna of known gain of 10 dBi at its operating frequency receives the power of P_s in its boresight when a transmit antenna transmits a power P_t at the far-zone of each other. With the same polarization as the standard antenna, the antenna under test receives the power of $0.5P_s$ in its boresight when the power transmitted by the transmitter is kept P_t . So, the boresight gain of the antenna under test is

- (a) 7 dBi
- (b) 8 dBi
- (c) 10 dBi.
- (d) 12 dBi.
- (e) 13 dBi.

$$10 \times \frac{0.5P_s}{P_s} = 10 + (-3) = 7$$

(3 marks)

8. If the dipole of directivity of 3.0 dBi has the radiation resistance of 40Ω and the material ohmic loss of 3Ω . The dipole antenna is fed by a $50\text{-}\Omega$ feeding line without any impedance matching network between the dipole and feeding line. So, the radiation efficiency and realized gain are, respectively

- (a) 93.0% and 3.0 dBi
- (b) 92.5% and 3.0 dBi.
- (c) 93.0% and 2.67 dBi.
- (d) 92.5% and 2.67 dBi.

$$\epsilon_r = \frac{40}{43} = 93\% \quad \epsilon_a = 92.5\%$$

$$T = \frac{Z_A - Z_0}{Z_A + Z_0} = \frac{43 - 50}{43 + 50} = 0.075$$

$$1 - |T|^2 = 0.9943$$

$$G = D \cdot \epsilon_a = 3 \times 92.5\% = 2.775 \text{ dBi} \approx 2.67 \text{ dBi}$$

(3 marks)

9. The input impedance of an HF RFID tag antenna (coil) is designed for

- (a) matching to a standard feed-line impedance of 50Ω .
- (b) matching to the output impedance of the RFID tag chip used.
- (c) matching to the conjugate output impedance of the RFID tag chip used.
- (d) high radiation efficiency of the coil antenna.
- (e) high gain of the coil antenna

(3 marks)



10. Consider a full-wave dipole fed by a voltage source at its center as shown in Figure Q1 (a). There are three electric current distributions as shown in Figures Q1(b-d). Which answer is correct?
- (a) The electric current distribution (b);
 - (b) The electric current distribution (c);
 - (c) The electric current distribution (d);

(3 marks)

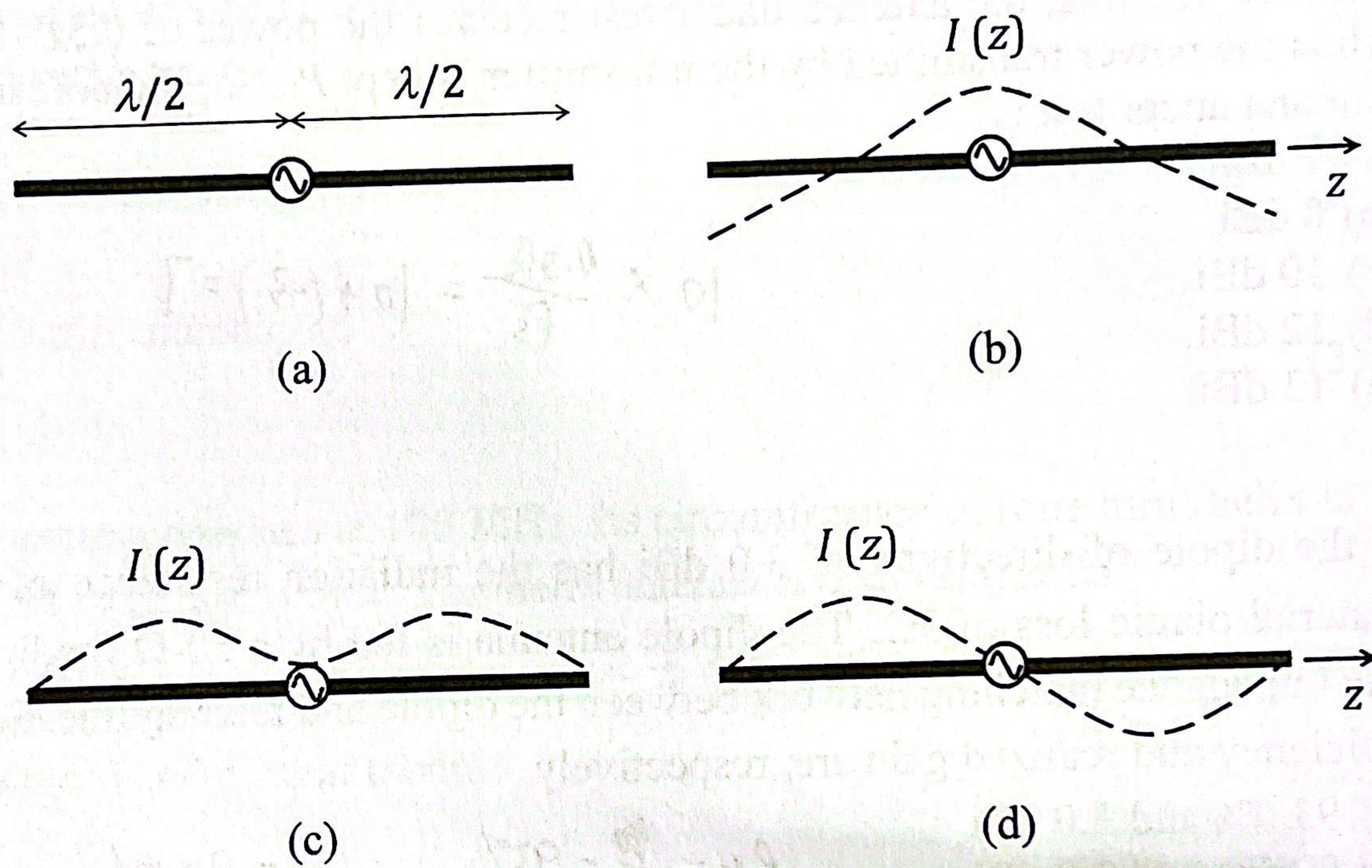


Figure Q1



Q2. (Total 30 marks)

1. A horn antenna with an aperture of $0.2 \times 0.2 \text{ m}^2$ operating at 2.45 GHz. Its farfield zone distance is 1.42 m. (5 marks)
2. Radiated E-field in a farfield zone ($r \gg \lambda$) of an antenna is $\mathbf{E} = \mathbf{a}_\theta V \frac{e^{-j\beta r}}{r} |\cos\theta| \text{ V/m}$. Its half-power beamwidth is 90 degrees. (5 marks) $|\cos\theta|^2 = 0.5$
 $\theta_1 = 45^\circ \theta_2 = 135^\circ$
3. A reflector antenna has a physical area of aperture $A=100$ square wavelength, aperture efficiency $e_{ap}=0.7$, and antenna efficiency $e_{ant}=0.9$. Its realized gain at boresight: 28.99 dB (5 marks)
 $A_e = e_{ap} \times A = 0.7 \times 100 = 70 \lambda^2$ $G = \frac{4\pi}{\lambda^2} e_{ant} A_p$
4. Design an H-plane sectoral horn antenna of gain of 16 dBi when other losses are ignored. The narrow wall is 0.5λ high (λ : wavelength in free space) and aperture efficiency is $e_{ap}=0.5$. Find: (15 marks)
 a) Effective aperture area = 6.34 λ^2 ;
 b) Length of H-plane of the horn = 12.67 λ ;
 c) Optimum axis length of horn = 10.35 λ ;
 d) Estimated HPBW = 6.16 degree; and
 e) Polarization if the horn is fed by rectangular waveguide operating at TE_{10} mode: Linear

$$G = 16 \text{ dBi} = 39.81 = \frac{4\pi}{\lambda^2} e_{ant} A_p$$

$$e_p = 0.5 \quad A_p = 4 \times 0.5 \lambda$$

$$\Rightarrow L_H =$$

$$A = 100 \lambda^2$$

$$A_e = 70 \lambda^2 = \frac{\lambda^2}{4\pi}$$

$$G_r = 70 \lambda^2 \times \frac{4\pi}{\lambda^2} = 280\pi$$

$$280\pi \times 0.9 = 252\pi = 28.99 \text{ dB}$$

$$\frac{4\pi}{\lambda^2} \times 0.5 \times L_H^2 \times 0.25 = 39.81$$

$$L_H^2 = \frac{39.81}{4\pi \times 0.5 \times 0.25} = 25.34$$

$$\frac{4\pi}{\lambda^2} \times 0.5 \times L_H \times 0.5 = 39.81 \lambda$$

$$L_H = 12.67 \lambda = \sqrt{3} R$$

$$A_p = 6.34 \lambda^2$$

$$L_E =$$

$$\frac{4\pi}{\lambda^2} \times 0.5$$

$$R = 53.51 \lambda$$

$$3R = 160.53 \lambda$$



Q3. (Total 20 marks)

A square array consisting of four isotropic antenna elements 1-4 is configured as shown in Fig Q3. All the elements are equally excited with a magnitude of I (A) and no phase difference. The uniform antenna spacing is $d=\lambda/2$ (one quarter wavelength) apart.

Hint:

The electrical far-field radiated at an observation point (at a distance $r \gg \lambda$ from the origin) by an isotropic source located at the origin carrying the excitation current magnitude of I :

$$\propto I \frac{e^{-jkr}}{r}.$$

- Give all the approximate distances r_1 and r_2 in terms of r_0 , spacing d and angle θ at the far-field observation point (5 marks)
- Find and simplify the expression of the array factor in the y-z plane for this array. (5 marks)
- Find angles (in degree) for all the possible nulls in the array factor (5 marks)
- Find angles (in degree) of all the maxima and minima in the array factor (5 marks)

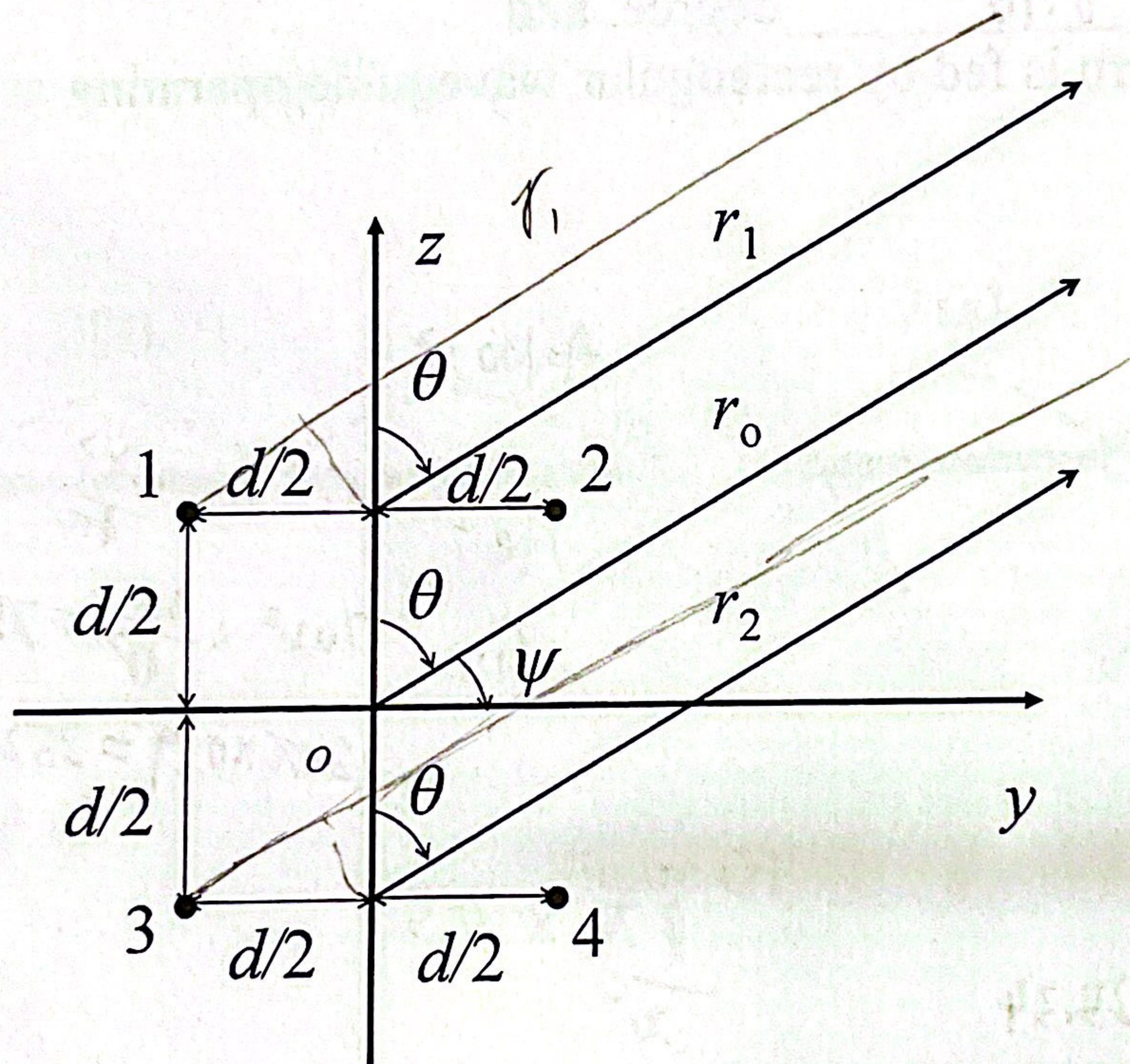


Figure Q3

$$r_1 = r_0 - \frac{d}{2} \cos \theta \quad r_2 = r_0 + \frac{d}{2} \cos \theta$$

$$\theta = 45^\circ \quad \sin \theta = \cos \theta = \frac{\sqrt{2}}{2}$$

$$d = \frac{\lambda}{2} \quad kd = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$$

$$\frac{kd \sin \theta}{2} = \frac{\pi}{2} \sin \theta = (2n+1) \frac{\pi}{2}$$

$$\sin \theta = 2n+1 \quad \sin \theta = \pm 1 \quad \theta = 90^\circ$$

$$\frac{kd \cos \theta}{2} = \frac{\pi}{2} \cos \theta = (2n+1)$$

$$\cos \frac{\pi \sin \theta}{2} \cos \frac{\pi \cos \theta}{2}$$

$$\frac{kd \sin \theta}{2} = \frac{\pi}{4}$$

$$\frac{\pi}{4} \pm n\pi \quad \frac{5}{4} \sim \quad -\frac{3}{4}$$

$$\frac{\pi}{4} \sin \theta = \frac{\pi}{4} \quad \sin \theta = \frac{1}{2}$$

Question 4 in next page



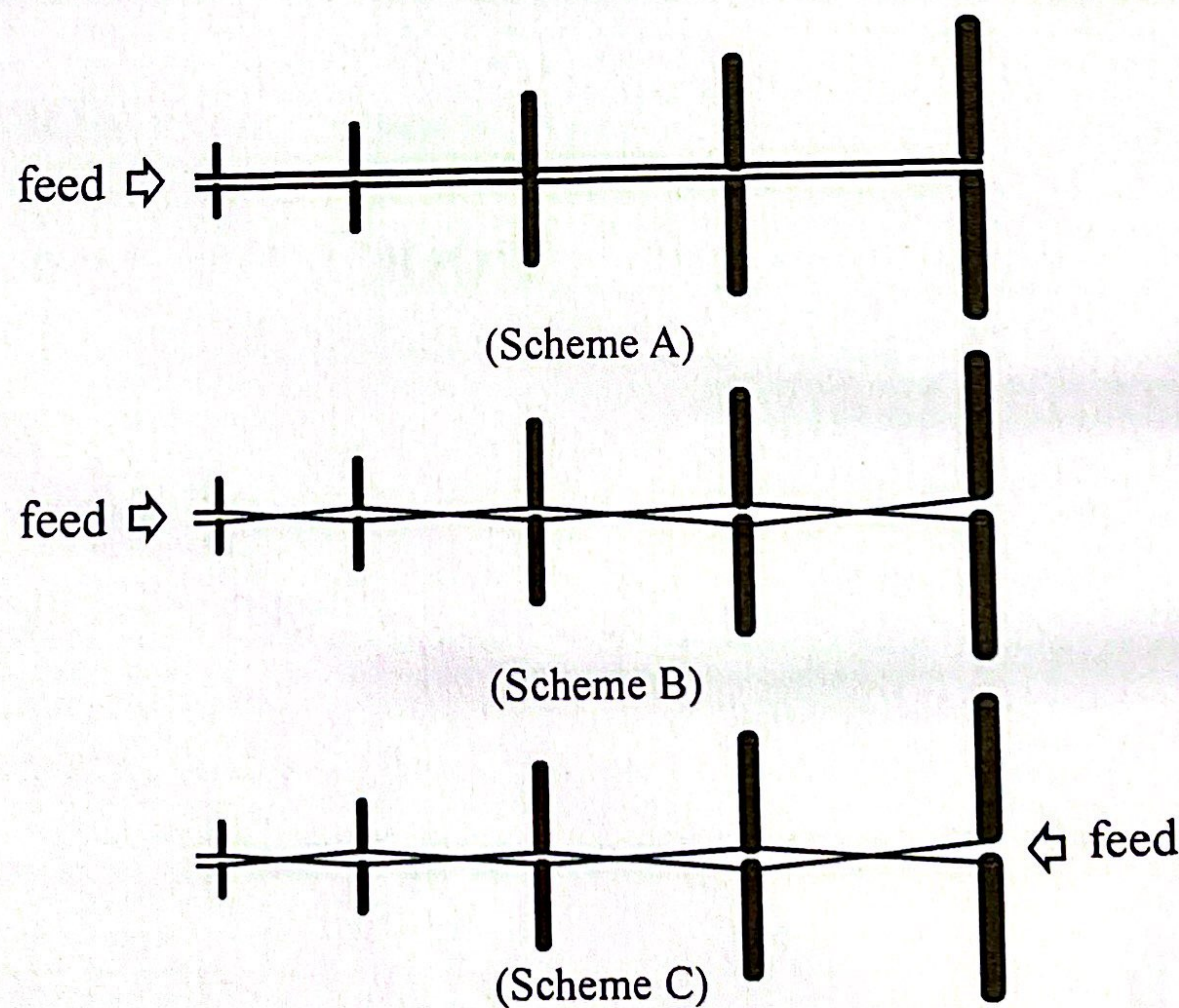
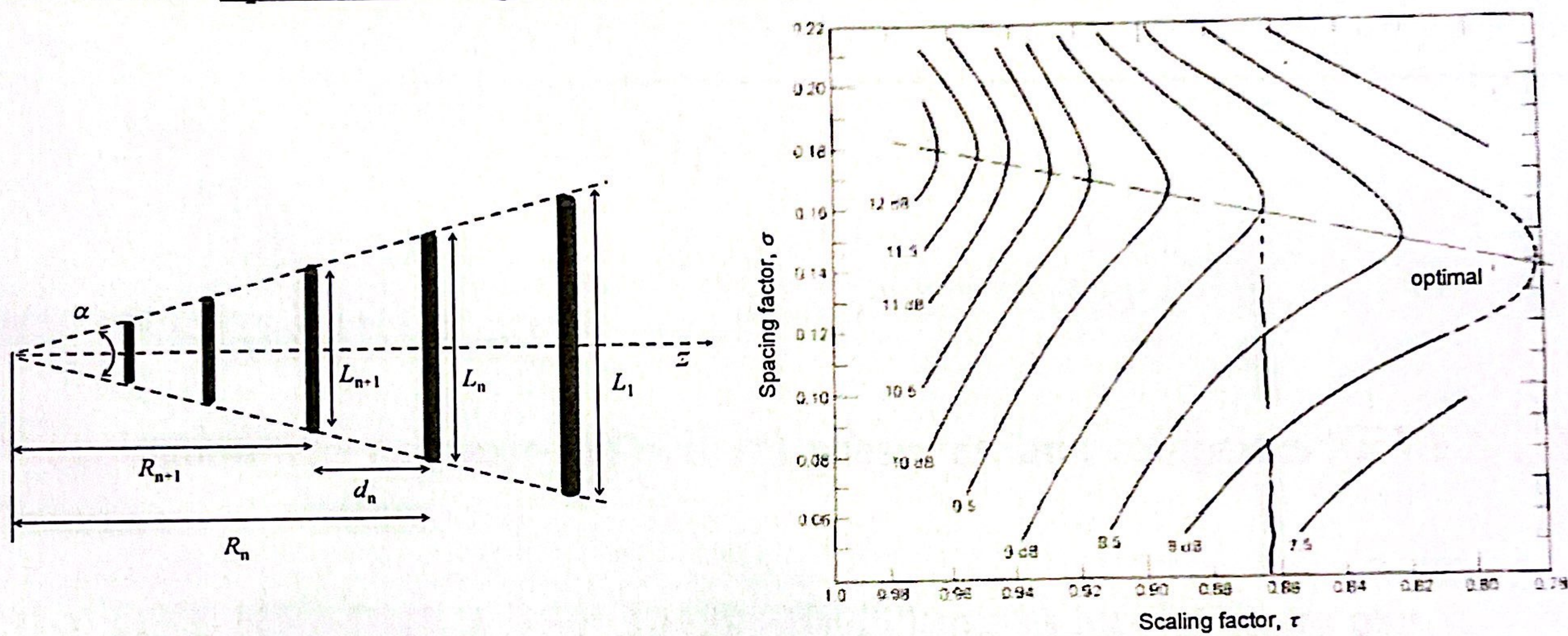
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Q4: (Total 20 marks)

A log-periodic dipole antenna (Figure Q4) with optimum parameters is designed to cover the frequency range of 47–80.6 MHz. Suppose the desired gain is 9dBi. Ignore the variation of feed gap, s_n .

- Determine the spacing and scaling factors σ and τ for the optimum design (5 marks)
- Determine the minimum number of the dipoles, N for the realized gain over the desired frequency range (5 marks)
- Tabulate the lengths of all dipoles, L_n in meter and all the element spacing, d_n in meter in Table Q4 (on Answer Book) (5 marks)
- Select the most suitable feeding scheme A, B, or C for the high-gain design (5 marks)

Hint: to determine optimum design factors from the attached figure.



$$(a) \tau = 0.862 \quad \sigma = 0.158$$

Figure Q4

Table Q4 List of L_n and d_n

n	1	2	3	4	5	6
L_n (m)						
d_n (m)						

END OF PAPER



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